Hrushikesh Vazurkar

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Personal Profile

Highly motivated and adaptable data professional with a proven ability to build strong relationships through effective communication. Well-versed in agile methodology, software development, big data and machine learning. Pursued a master's degree from the University of Edinburgh to deepen expertise in data analysis and modelling, with a goal of playing a pivotal role in enhancing your company's software lifecycle and data workflows.

Education

University of Edinburgh — Edinburgh, UK
MSc. in Statistics with Data Science ***(2:1 Expected)*** — *Aug 2023 – Sep 2024*

National Institute of Technology — Nagpur, India
BTech. in Computer Science and Engineering ***(8.32/10)*** — *Aug 2017 – Jun 2021*

Work Experience

Associate Data Engineer — Aug 2021 – Aug 2023
Lowe's Home Improvement

- Enabled Hadoop 2.0 decommissioning by migrating 5 legacy data pipelines to Airflow in Hadoop 3.0 environment.
- Delivered BI dashboards on Apache Superset to highlight metrics regarding products returned by customers, supporting stakeholder negotiations with product vendors.
- Contributed to building ETL data pipelines to augment the above dashboard – including extracting data from relevant upstream sources in Apache Hive to Apache Druid, and maintaining correspondence with downstream users.
- Spearheaded the development and deployment of a PoC application in Spring Boot with PostgreSQL for generating product-specific tax codes – including unit and integration testing with Junit and Mockito, code reviews and sprint retrospectives.

Summer Intern — May 2020 – Jun 2020
Fidelity Investments

- Delivered a comprehensive research paper on the comparative analysis of BERT-based LLMs and conventional models, providing a solid groundwork for model selection for other NLP related use-cases at the Asset Management Group.
- Open-sourced our codebase on Fidelity GitHub page as a plug-and-play Python package for ease of use of LLMs for text classification (https://github.com/fidelity/classitransformers).
- Achieved state-of-the-art accuracy on Yelp 2013 (69%) and Financial Phrase bank (88.2%).

Projects

Insurance Complaint Analysis | *Python, PyTorch, Scikit-learn, BERT, LLM*
MSc Dissertation Project 2

- Uncovered temporal and insurance-wise patterns of customer complaints through exploratory data analysis.
- Developed a predictive FOS model using Distil-BERT with down-sampling, to replicate the **Financial Ombudsman Service** decisions for customer complaints, achieving 84% accuracy.
- Prepared a comprehensive report providing timeline of overall complaint spikes, and strategies for scrutinizing insurance products based on both complaint upheld likelihood and the FOS model prediction.

Unsupervised Time-series Modelling | *PyTorch, Scikit-learn, Statsmodels*
MSc Dissertation Project 1

- Delivered an unsupervised model for the **Lloyd's Bank** to identify risky transactions based on an hourly time-series employee spending dataset.
- Performed exploratory data analysis and time-series diagnostic tests to validate attributes like stationarity, inter-employee correlation and autocorrelation for feature and model selection.
- Achieved optimal results for Isolation Forest with F1 scores of 0.94 and 0.23 for identifying non-risky and risky events.

Skills

Programming languages – Java, Python, R, JavaScript, SQL; **Libraries and Frameworks –** Spring Boot, React.js; **CI/CD –** Git, Docker, Jenkins; **Tools –** Apache Superset, Apache Kafka

Publications

Chhipa, M., Vazurkar, H. M., Kumar, A. and Mishra, M. (2020). Comparative study of language models on cross-domain data with model agnostic explain-ability. https://doi.org/10.48550/arXiv.2009.04095